# Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

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For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

## 0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I plan to use loops, if statements, lists, classes and make a graph at the end. I want to improve my ability to make loops with more difficult patterns and make graphs with lists I made.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I expect the if statements and lists work to be the easiest. We have spent a long time doing that and I feel very confident in my ability to use them. I think the drafter UI and dataclasses will be the most difficult. I still struggle with drafter and sometimes struggle to access dataclasses.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think creating the routes and look of drafter will take the longest. I should implement the helper functions and basic operations so I can build the drafter around it.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions I have are my physics and math exams coming up. I will need to spend time studying for both. I will get ahead on my work for this week to free more time for this project. I can talk to a TA, a friend or LLM(with documentation) if I need help. I do not currently have any questions.

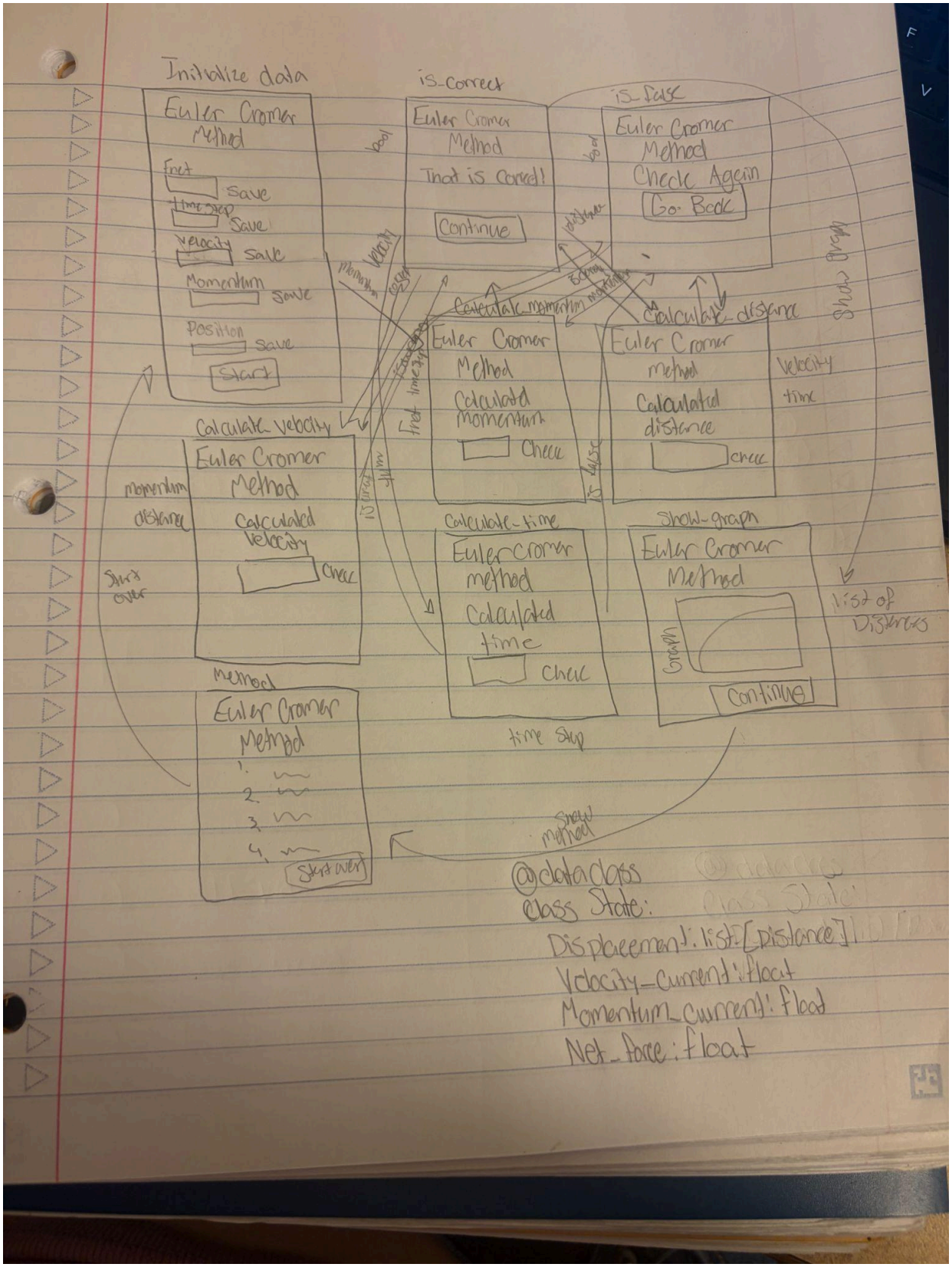
# 1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Initialize data
Euler Cromer Method
Fnet
Save
time step
Save
velocity
Save
Momentum
Save
Position
Save
Start
is_correct
Euler Cromer Method
That is correct!
Continue
is_false
Euler Cromer Method
Check Again
Go Back
bool
bool
distance
Calculate_momentum
Calculate_distance
Euler Cromer Method
Calculated momentum
Check
Euler Cromer Method
Calculated distance
check
Velocity
time
Calculate_velocity
Euler Cromer Method
Calculated Velocity
Check
momentum
distance
Start over
Calculate_time
Euler cromer method
Calculated time
Check
Show_graph
Euler Cromer Method
graph
Continue
list of Displacements
Method
Euler Cromer Method
1.
2.
3.
4.
Start over
time step
Show method
@dataclass
class State:
Displacement: list[distance]
Velocity_current: float
Momentum_current: float
Net_force: float

# 2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

| **Requirement** | **Implementation** |
|---|---|
| **Purpose/Functionality**<br><br>A 1-2 sentence description of the purpose or functionality of your site | **The purpose of my site is to teach students how to do the euler cromer method and create a visual to show the movement of the particle.** |
| **State Dataclass Attributes**<br><br>Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass.<br><br>Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site. | class State:<br>Displacement:list[Distance]<br>Velocity_current: float<br>Momentum_current:float<br>Net_force:float<br><br>The displacement is a list of distances that can be added to get how much the object has moved<br>Velocity_current is the updated velocity needed for calculations. As the object moves it will be modified<br>Momentum_current is what the current momentum is and will be need to be updated with time and Net_force<br>Net_force will start the process, it will not be changed but will need to be saved for calculations |
| **Second Dataclass**<br><br>Your second dataclass, including its attributes, their types, and how they will be meaningfully modified | class Calculations:<br>Distance:float<br>Time: float<br>The distance is what is calculated in every step. Every instance will go into the list for modification. |

| | |
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| (if they are modified) | The time will not be modified but will need to be set by the user and used for all calculations. |
| **Routes** (minimum 7)<br><br>The names of your route functions | index<br>Calculate_momentum<br>Update_velocity<br>Update_distance<br>Update_time<br>Create_graph<br>Show_method |
| **Pages** (minimum 5)<br><br>A brief description of the pages in your site | For each step I want the user to be able to check their work. I am making a page that says if something they type in correct and a page that says something is incorrect. I page to input original data like initial velocity, momentum, distance and time. A page each for inputting the velocity, distance, momentum and net force they calculated, a page that shows the graph and a page that shows the full process for any problem for future reference. |
| **User Input** (minimum 3)<br><br>Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes<br><br>For more information on these input types, see Drafter Documentation. | They will have textboxes for inputting time, net force, velocity, momentum and displacement |
| **If Statements** (minimum 3)<br>Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site | For each time a user inputs something into the site, it will run through an if statement that checks if the input data is equal to the calculated data and returns a page telling them to check again or continue. This will happen 5 times. This is meaningful because it achieves the goal of teaching a student how to use the method with any data and can be used infinitely |

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| | many times. |
| **Loop** (minimum 1)<br>This loop must iterate over the list/dict of dataclass instances in your State attribute.<br>Description of this loop and why it is meaningful to the site | I will use a loop to access my list of distances and create a graph. I am also considering using a loop to make the data smaller so it fits better on my graph. This is meaningful because it creates a visual that shows an asymptote if the object is moving close to the speed of light which is very important for understanding projection of objects. |

# LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

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|---|
| *I did not use an LLM* |

# Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.